

Capstone Project Plan: Game Achievement & Difficulty Tracker

Stack: Spring Boot, React, TypeScript, Spring Security

Overview

A web app for tracking game backlogs with a focus on achievements. Users link their Steam account to import their library and achievement progress, rate the difficulty of individual achievements and games overall, and see a global leaderboard of the hardest games and individual achievements as voted by the community.

MVP

User accounts

Registration/login with Spring Security, JWT, and bcrypt-hashed passwords.

Steam account linking

User provides their Steam profile URL/vanity name; backend resolves it to a SteamID64 and imports their owned games via the Steam Web API.

Manual game entry

Users can also add games not pulled from Steam, using IGDB or RAWG for metadata/cover art.

Library management

Add games to library, set status (backlog/playing/completed/dropped), leave notes, rate the game overall.

Achievement tracking

Per game, fetch the achievement schema and the user's unlock status, display as a checklist with icons and unlock dates.

Difficulty ratings

Users rate difficulty at two levels: individual achievements and the game as a whole.

Global difficulty leaderboard

Games ranked by average user-submitted difficulty rating, with a minimum-vote threshold so low-sample games don't skew the top of the list.

Stretch Goals

Background library sync (JobRunr)

Replace per-game on-demand achievement fetching with a background job that syncs a user's entire library at once, respecting Steam API rate limits, with sync status the frontend can poll.

Community-aggregated achievement difficulty

Average difficulty rating per individual achievement, shown alongside the user's own rating.

OAuth login (Google or GitHub)

Added as a second login option alongside username/password.

Social features

Friend requests, viewing a friend's library, optionally a friends-scoped leaderboard view.

Live activity feed (WebSockets)

Real-time feed of friend activity.

Docker

Containerize the app and database for easier local setup and deployment